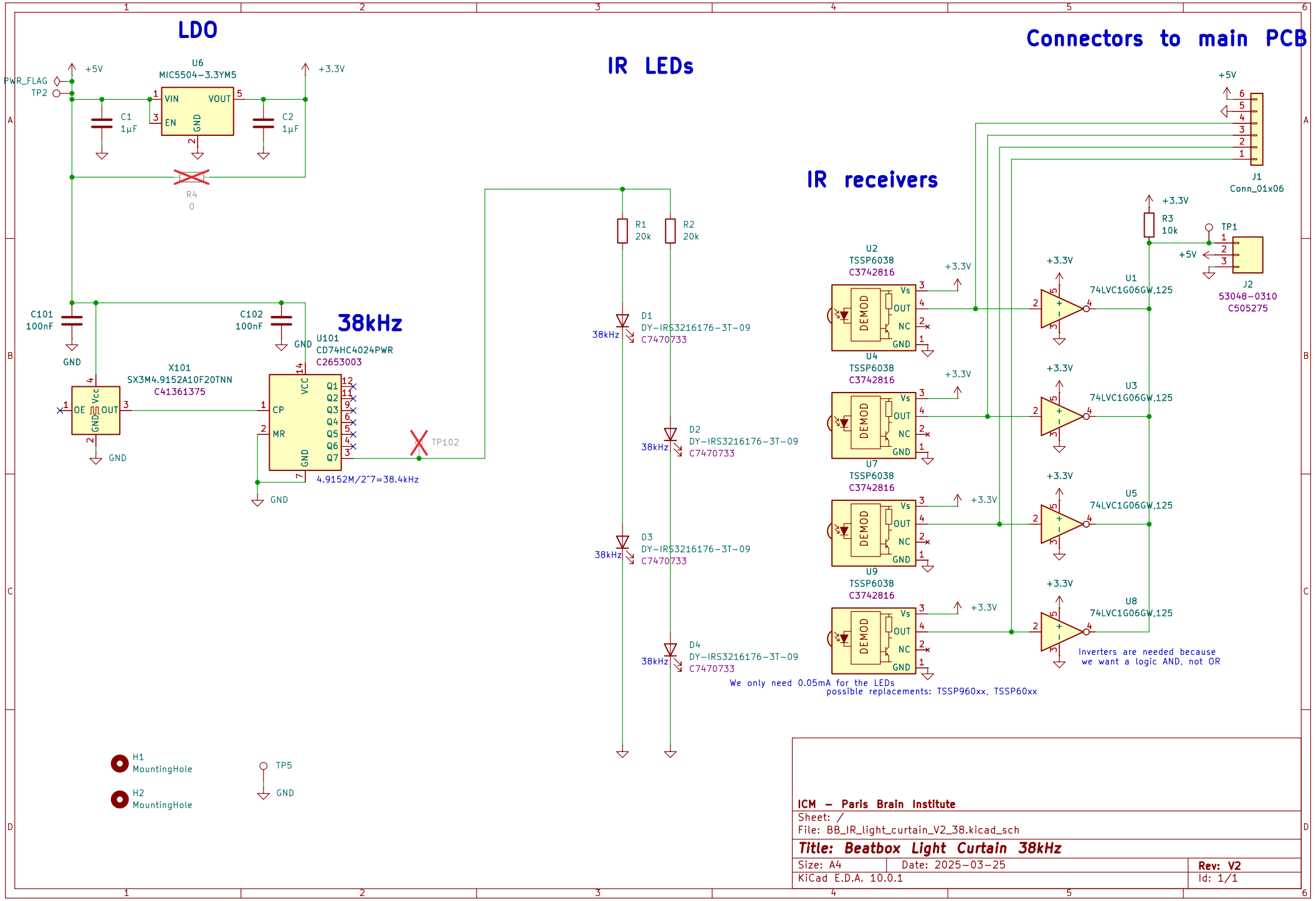

LDO
IR LEDs
Connectors to main PCB
IR receivers
38kHz
PWR_FLAG
TP2
+5V
U6
MIC5504-3.3YM5
VIN
VOUT
EN
GND
+3.3V
C1
1µF
C2
1µF
R4
0
C101
100nF
C102
100nF
GND
X101
SX3M4.9152A10F20TNN
C41361375
OE
Vcc
OUT
U101
CD74HC4024PWR
C2653003
CP
MR
Q1
Q2
Q3
Q4
Q5
Q6
Q7
TP102
4.9152M/2^7=38.4kHz
R1
20k
R2
20k
D1
DY-IRS3216176-3T-09
C7470733
38kHz
D2
DY-IRS3216176-3T-09
C7470733
D3
DY-IRS3216176-3T-09
C7470733
D4
DY-IRS3216176-3T-09
C7470733
We only need 0.05mA for the LEDs
possible replacements: TSSP960xx, TSSP60xx
U2
TSSP6038
C3742816
U4
TSSP6038
C3742816
U7
TSSP6038
C3742816
U9
TSSP6038
C3742816
DEMOD
Vs
OUT
NC
GND
U1
74LVC1G06GW,125
U3
74LVC1G06GW,125
U5
74LVC1G06GW,125
U8
74LVC1G06GW,125
Inverters are needed because
we want a logic AND, not OR
R3
10k
TP1
J1
Conn_01x06
J2
53048-0310
C505275
H1
MountingHole
H2
MountingHole
TP5
ICM – Paris Brain Institute
Sheet: /
File: BB_IR_light_curtain_V2_38.kicad_sch
Title: Beatbox Light Curtain 38kHz
Size: A4
Date: 2025-03-25
Rev: V2
KiCad E.D.A. 10.0.1
Id: 1/1